

## DAY-3

### Tasks

1.

List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

| BIOS/POST error message                  | Usually indicates                                             |
|------------------------------------------|---------------------------------------------------------------|
| 1. CMOS Checksum Error                   | CMOS data is corrupted or the CMOS battery is weak/dead       |
| 2. Keyboard Error or No Keyboard Present | Keyboard is disconnected, faulty, or not detected             |
| 3. Disk Boot Failure                     | Boot drive is missing, faulty, or the boot order is incorrect |
| 4. Memory Error / Memory Size Mismatch   | RAM may be loose, faulty, or incorrectly installed            |
| 5. CPU Fan Error                         | CPU fan is disconnected, not working, or spinning too slowly  |

2.

Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

For a traditional AMI BIOS, 3 beeps typically indicate a **base 64 KB memory failure**. In simple terms, the BIOS is detecting a problem with the system RAM.

#### Troubleshooting step 1: Reseat the RAM

1. Turn off the PC and disconnect the power.
2. Remove the RAM module.
3. Check the RAM and slot for dust or visible damage.
4. Reinsert the RAM firmly.
5. Start the PC again.

#### Troubleshooting step 2: Test the RAM

- If there are multiple RAM sticks, test them **one at a time**.
- Try a different RAM slot if available.
- If one stick works and another produces the beeps, the faulty RAM module is likely the problem.

Note: Beep codes can vary between BIOS versions and motherboard manufacturers, so the motherboard's manual should be checked for the exact code.

### 3.

Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

#### Steps

1. Shut down the PC.
2. Turn it on.
3. Immediately press the BIOS key repeatedly.
  - Common keys: F2, Delete, F10, or Esc.
4. The BIOS/UEFI Setup screen will appear.
5. Take a screenshot or photograph of the main BIOS screen.
6. Look for information such as:
  - Motherboard manufacturer
  - BIOS/UEFI version
  - BIOS date
  - CPU information
  - RAM information

### 4.

You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.  
  
**Hint:** Think about the CMOS battery and BIOS settings reset.

The message:

**"CMOS checksum error – Defaults loaded"**

means that the BIOS detected that the stored CMOS configuration does not match the expected checksum. As a result, the system has loaded the default BIOS settings. A weak/dead CMOS battery is a common cause, although corrupted BIOS settings or an unsuccessful BIOS update can also cause it.

#### **How to resolve it**

1. Restart the computer and enter BIOS Setup.
2. Choose **Load Defaults / Optimized Defaults**.
3. Check the date and time.
4. Check important settings such as boot order.
5. Save the settings and restart.
6. If the error comes back repeatedly, shut down the PC and disconnect power.
7. Replace the CMOS/RTC battery, commonly a CR2032 in desktop PCs.
8. Start the PC and enter BIOS again.
9. Set the correct date, time, and boot settings.
10. Save and restart.

If the error continues even after replacing the battery and restoring the defaults, the BIOS firmware or motherboard may require further troubleshooting.